

MAINTENANCE OPERATIONS CLIENT

Power BI Requirements

Maintenance Operations Analytics & Reporting Automation

Document Control	Detail
Engagement	Aug-Nov 2024
Primary Stakeholder	Maintenance Manager
Solution	Power BI dashboard built from scratch
Data Scope	Approximately 10,000 historical records across 20 teams
Business Outcome	Approximately 3-4 hours of weekly manual reporting effort eliminated
Status	Portfolio requirements based on confirmed project scope

Scope boundary: Exact historical dashboard page names, layout, refresh method and original DAX implementation are not retained. Proposed portfolio page structure is identified as reconstruction.

1. Solution Objective

Provide the Maintenance Manager with a consolidated Power BI view of maintenance performance, outstanding work and operational exceptions, replacing recurring manual reporting with reusable analytical reporting.

2. Core Reporting Requirements

ID	Requirement	Priority
PBI-01	Display overall maintenance Completion Rate using the governed KPI definition.	Must
PBI-02	Display completed, open and pending maintenance work.	Must
PBI-03	Identify overdue work using Due Date < Current Date AND Status != Completed.	Must
PBI-04	Support analysis of average completion time using the documented portfolio calculation.	Must
PBI-05	Provide priority breakdown and visibility into high-priority pending work.	Must
PBI-06	Support team-wise completion and team performance comparison.	Must
PBI-07	Support technician-level performance analysis where attribution is valid.	Must
PBI-08	Provide backlog visibility using one explicitly documented reconstructed backlog definition.	Must
PBI-09	Support analysis of workload imbalance.	Should
PBI-10	Support recurring maintenance-category analysis.	Should
PBI-11	Support asset/equipment breakdown and equipment-related pattern analysis.	Should
PBI-12	Allow users to filter relevant views using available operational dimensions.	Must

3. KPI Requirements

KPI / View	Requirement	Evidence Status
Completion Rate	Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100	Confirmed historical rule
Completion Rate result	92% overall completion rate may be presented as the retained engagement finding	Confirmed finding
Overdue	Due Date < Current Date AND Status != Completed	Confirmed historical rule
Completed / Open / Pending	Use standardized status mapping	Exact historical mapping not retained
Average Completion Time	Use documented rebuilt date convention	Exact historical formula not retained
Backlog	Use one explicit reconstructed definition	Exact historical definition not retained
Team / Technician	Use governed underlying measures rather than an invented composite score	Confirmed analytical areas

92% interpretation: The 92% completion rate is an observed analytical result, not a project-caused performance improvement.

4. Required Filters & Dimensions

Dimension	Reporting Use
Team	Team performance and workload analysis
Technician	Technician performance where attribution is valid
Status	Completed/open/pending analysis
Priority	Priority and critical pending-work analysis
Category	Recurring maintenance-category analysis
Asset / Equipment	Equipment-level maintenance analysis
Location	Operational slicing where appropriate
Dates	Historical/time-based analysis where valid

5. Proposed Portfolio Dashboard Structure

Reconstruction note: The following page structure is proposed for the portfolio. It is not claimed as the historical dashboard layout.

Page	Purpose	Key Content
Executive Maintenance Overview	Management-level operational position	Completion Rate, work status, overdue/backlog visibility, priority context
Work Order & Completion Analysis	Work execution and timeliness	Completed/open/pending, overdue, completion time
Team & Technician Performance	Operational performance comparison	Team completion, technician measures, workload distribution
Backlog, Priority & Asset Analysis	Exception and pattern analysis	Backlog, high-priority pending, recurring categories, asset/equipment patterns

6. Interaction Requirements

ID	Requirement
INT-01	Filters must update relevant KPI cards and analytical visuals consistently.
INT-02	Equivalent measures must reconcile across pages under equivalent filter context.
INT-03	Users must be able to move from overall performance to team, technician, priority, category and asset/equipment detail where supported.
INT-04	Unknown/missing categories must not be silently reassigned to valid operational values.
INT-05	Overdue views must use the confirmed overdue rule consistently.

7. Data & Quality Requirements

Requirement	Control
Duplicate control	Prepared data must prevent duplicate records from inflating counts/rates.
Team consistency	Team names must be standardized before team-level reporting.
Status consistency	Status values must be standardized before completion/open/pending/overdue reporting.
Date validity	Due and completion-related date fields must be valid for date-based measures.
Missing values	Unsupported operational values must not be fabricated.
Confidentiality	Public portfolio outputs must use sanitized/synthetic identifiers and values where required.

8. Functional Acceptance Criteria

ID	Acceptance Criterion
AC-01	Completion Rate reconciles to the governed formula.
AC-02	The retained 92% result is labeled as an observed completion KPI, not improvement.
AC-03	Every overdue record satisfies the confirmed overdue rule.
AC-04	Completed records do not appear as overdue.
AC-05	Team-level completion uses the same completion definition as the overall KPI.
AC-06	Backlog uses one documented reconstructed definition consistently.
AC-07	High-priority pending logic uses documented priority and status mappings.
AC-08	Filters produce consistent results across relevant visuals.
AC-09	No illustrative/sanitized portfolio value is represented as historical client data.

9. Traceability

Requirement Area	BRD	User Story	KPI / Analysis
Completion & status	BR-02 to BR-05	US-01, US-02	Completion Rate, completed/open/pending
Overdue	BR-06	US-03	Overdue Work Orders
Completion time	BR-07	US-04	Average Completion Time
Priority	BR-08, BR-13	US-05	Priority, high-priority pending
Team	BR-09, BR-12	US-06	Team completion, workload
Technician	BR-10	US-07	Technician performance
Backlog	BR-11	US-08	Backlog
Category / asset	BR-14, BR-15	US-09	Recurring categories, asset/equipment

10. Delivery Requirements

Requirement	Confirmed Position
Dashboard	Power BI dashboard built from scratch
Delivery	Live walkthrough and presentation
Supporting material	Documentation and training
Reporting benefit	Approximately 3-4 hours of weekly manual reporting effort eliminated
Refresh method	Not retained; no historical refresh mechanism should be claimed

11. Portfolio Implementation Boundary

The Data Model & DAX Measures document may implement these requirements as a reproducible portfolio solution. Exact historical DAX, Power Query logic, schema design and dashboard layout must not be presented as original client implementation unless additional evidence is supplied.

Public dashboard rule: Any dashboard mockup using synthetic values should be labeled: SANITIZED PORTFOLIO MOCKUP - ILLUSTRATIVE VALUES, NOT CLIENT DATA.